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Speech recovery based on the linear canonical transform ☆

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Abstract

As is well known, speech signal processing is one of the hottest signal processing directions. There are exist lots of speech signal models, such as speech sinusoidal model, straight speech model, AM–FM model, gaussian mixture model and so on. This paper investigates AM–FM speech model by the linear canonical transform (LCT). The LCT can be considered as a generalization of traditional Fourier transform and fractional Fourier transform, and proved to be one of the powerful tools for non-stationary signal processing. This has opened up the possibility of a new range of potentially promising and useful applications based on the LCT. Firstly, two novel recovery methods of speech based on the AM–FM model are presented in this paper: one depends on the LCT domain filtering; the other one is based on the chirp signal parameter estimation to restore the speech signal in LCT domain. Then, experiments results are presented to verify the performance of the proposed methods. Finally, the summarization and the conclusion of the paper is given.

Keywords: Linear canonical transform; Chirp signal; The AM–FM model; Speech; Signal reconstruction

1. Introduction

As is known, a speech signal is a kind of typical non-stationary signal whose frequency components vary with time, which is consisting of vowels, consonants and transient parts. It is very important to detect its spectral components in many speech processing applications. The traditional tool for this estimation is spectral analysis by means of Fourier transform, however, this is a tool for stationary signals processing only and it simply indicates the global frequency components, but does not tell us when they occurred. This defection has seriously hindered the development of the speech signal processing. Focus on the speech signal processing, many time–frequency analysis tools have been proposed, for example, the short-time Fourier transform (STFT), the wavelet transform (WT), the fractional Fourier transform (FRFT) (Yin et al., 2008).

The speech signal, as a special kind of signal by some harmonic structures consisted, based on the mechanism of the production of speech, people has proposed some speech models, such as speech sinusoidal model, straight speech model, gaussian mixture model and so on (Yin et al., 2008; Teager and Teager, 1989). Besides, the speech also can be seen as a special kind of multicomponent chirp signals (Bovik et al., 1993). A chirp is an important signal in which the frequency increases ('up-chirp')or decreases ('down-chirp') with time. Chirp signal is a typical non-stationary signal, which widely appears in systems such as communication, radar, sonar, biomedicine and earthquake exploring system and also be used as a signal model for a lot of natural phenomena. Therefore, it is very important for chirp signal processing. The detection, the parameters estimation and the time–frequency representation for the multicomponent chirps are still an important research topic. In 1980s Teager and Teager (1989) discoveres by experiments that vortices could be the secondary source to excite the channel and produce the speech signal. Therefore, speech should be composed of the plane-wave-based linear part and a vortices-based non-linear part. According

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to such theory, Dimitriadis and Maragos (2005), Bovik et al. (1993), Santhanam and Maragos (2000), and Huang (1998) proposed an AM–FM modulation model for speech analysis, synthesis and coding. During the last thirty years, the four milestones among the best practices methods associated with the speech detection and recovery are MESA (Bovik et al., 1993), PASED (Santhanam and Maragos, 2000), Hilbert Huang Transform (Huang, 1998) and Gianfelici Transform (Gianfelici et al., 2007).

The paper Bovik et al. (1993) develops a multiband or wavelet approach for capturing the AM–FM components of modulated signals immersed in noise. It is demonstrated that the performance of the energy operator/ESA approach is vastly improved if the signal is first filtered through a bank of bandpass filters, and at each instant analyzed using the dominant local channel response. In the paper Santhanam and Maragos (2000), authors present a nonlinear algorithm for the separation and demodulation of discrete-time multicomponent AM–FM signals, it avoids the shortcomings of previous approaches and works well for extremely small spectral separations of the components and for a wide range of relative amplitude/power ratios. The paper Gianfelici et al. (2007) developes another new method for analyzing nonlinear and non-stationary signal processes. This decomposition method is adaptive, and, therefore, highly efficient. The proposed approach presented in the paper Gianfelici et al. (2007) is on the basis of a rigorous mathematical formulation, and the validity of this approach has been proven by some applications to both synthetic signals and natural speech.

In this paper, we introduce two novel methods to restore the speech signal with background noise based the AM–FM model associated with the linear canonical transform (LCT). The LCT is also known ABCD transform, generalized Fresnel transform (Aizenberg and Astola, 2006), Collins formula (Fan and Lu, 2006), generalized Huygens integral (Kostenbauder, 1990). It was proposed early in the seventies by Moshinsky and Quesne (2006) and Collins (1970) the special case with complex parameters was proposed by Bargmann (1961). LCT can be considered as a generalization of traditional Fourier transform and fractional Fourier transform. As same as the fractional Fourier transform, the LCT is used for analyzing Optical Systems and solving Differential Equations in the first. With 1990s' development of fractional Fourier transform, LCT has began to be taken seriously in the field of signal processing community. In such circumstances, we can analysis the signal in LCT domain, because from Pei and Ding (2002) and Sharma and Joshi (2008) we know that the one-dimensional LCT is linear integral transform of a three-parameter class, LCT can be more general and flexible than the Fourier transform as well as the fractional Fourier transform in some properties (Sharma and Joshi, 2008), and it can solve problems that cannot be dealt with well by the latter. It is shown in Tao et al. (2009), Li et al. (2012), Sharma and Joshi (2008), and Maragos etal. (1993) that the LCT is one of the powerful non-staionary signal processing tools, especially suitable for time-varying chirp signal processing. So it is most hopeful by use of LCT for this speech signal processing based AM–FM model. So far, LCT has many applications in signal processing community. For example, it has been applied to filter design, communication signals against multi-path and so on (Li et al., 2012; Sharma and Joshi, 2006; James and Agarwal, 1996).

The rest of the paper is organized as follows. In Section 2, the definition and some basic properties of LCT are briefly introduced. In Section 3, single chirp signal model is produced, then the AM–FM model of speech is described. In Section 4, based the AM–FM model we introduce two novel methods to restore the speech signal according to LCT. Section 5 presents the experimental results and discussion. Some conclusions and future research are given in Section 6.

2. Preliminary

When parameters (a, b, c, d) are real numbers, the LCT of a signal $f(t)$ defines as follows (Tao et al., 2009):

$$F_{(a,b,c,d)} = \begin{cases} \sqrt{\frac{1}{j2\pi b}} \int_{-\infty}^{\infty} K_{a,b,c,d}(u,t) f(t)\, dt & \text{if } b \neq 0 \\ \sqrt{d} \exp\left(\frac{jcd}{2} u^2\right) f(du) & \text{if } b = 0 \end{cases} \tag{1}$$

and $ad - bc = 1$, kernel function is defined as

$$K_{a,b,c,d} = \exp\left(\frac{jd}{2b} u^2\right) \exp\left(-\frac{j}{b} ut\right) \exp\left(\frac{ja}{2b} t^2\right), \tag{2}$$

Eq. 1 can be written as $F_{(a,b,c,d)} = L^{a,b,c,d}(f(t)) = L^A[f](u)$, the parameters

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

By the above definition it is proved in paper Moshinsky and Quesne (2006) that the LCT satisfies the additivity property of the parameters, that is

$$L^{a_2,b_2,c_2,d_2}[L^{a_1,b_1,c_1,d_1}(f(t))] = L^{e,f,g,h}(f(t)), \tag{3}$$

where

$$\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}.$$

Besides, the reversibility property is derived (Tao et al., 2009)

$$L^{d,-b,-c,a}[L^{a,b,c,d}(f(t))] = f(t). \tag{4}$$

When these parameters reduces to

$$(a, b, c, d) = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha),$$

then LCT becomes into the FRFT by a fixed phase factor multiplied (Moshinsky and Quesne, 2006):

$$L^{\cos\alpha,\sin\alpha,-\sin\alpha,\cos\alpha}(f(t)) = \sqrt{\exp(-j\alpha)} F^{\alpha}(f(t)). \tag{5}$$

When these parameters $(a,b,c,d)=(0,1,-1,0)$, LCT becomes into Fourier transform by fixed $\sqrt{-j}$ multiplied. When these parameters $(a,b,c,d)=(1,0,\tau,1)$, LCT reduces into chirp multiplication operator. Most of the classical concepts and theorems in the Fourier domain are generalized in the LCT domain, for example, the sampling theorem, the uncertainty principle and the Poisson sum formulae (Li and Xu, 2012; Li et al., 2007, 2009; Zhao et al., 2009; Tao et al., 2008; Stern, 2008). It has also found many advantages in real applications, such as filter design and signal synthesis, time–frequency analysis, electromagnetic wave propagation analysis, pattern recognition, communication signal modulation and multiplexing, encryption, etc. (Tao et al., 2009; Li et al., 2012; Sharma and Joshi, 2006; James and Agarwal, 1996; Stern, 2008). It is known that LCT is one of the most effective methods for non-stationary signals, speech is a kind of typical non-stationary signal. However, through my find no relevant paper is published about speech signal analyzing and processing by use of LCT method. So it is worthwhile and interesting to investigate speech signal processing in LCT domain. Later we will analysis a particular speech model by using LCT.

3. The AM–FM model of speech

In 1980s Teager and Teager (1989) discoveres by experiments that vortices could be the secondary source to excite the channel and produce the speech signal. Therefore, speech should be composed of the plane-wave-based linear part and a vortices-based non-linear part. According to such theory, Dimitriadis and Maragos (2005) proposes an AM–FM modulation model for speech analysis, synthesis and coding. The AM–FM model represents the speech signal as the sum of formant resonance signals each of which contains amplitude and frequency modulation (Maragos et al., 1993). In paper Maragos et al. (1993), we find that the AM–FM model of speech has a form of the multicomponent chirp signals model.

3.1. Chirp signal model

Assume a single chirp signal model:

$$f(t)=h\exp\left(j\frac{1}{2}a_2t^2+ja_1t+ja_0\right)+w(t), \tag{6}$$

where h is the signal amplitude parameter, a_2 is the frequency rate parameter of linear LFM signal, a_1 is known as the angular frequency parameter (also is known as the initial frequency or center frequency), a_0 is called the initial phase parameter, $w(n)$ is additive noise (Chew et al., 1994).

We consider that the received signal $y(n)$ consists of superimposed chirp signals of M embedded in noise, so the multi-component chirps signal model can be written as

$$y(t)=\sum_{i=1}^{M}h_i\exp\left(j\frac{1}{2}k_it^2+jb_it+jc_i\right)+w_1(t), \tag{7}$$

and the meaning of each component parameter is as same as the single chirp signal in (20).

3.2. AM–FM model of speech

In speech recognition, coding and synthesis, pitch and formants are always the most important parameters. In traditional speech processing methods, these parameters are considered as constant with a frame during 10–30 ms (Dimitriadis and Maragos, 2005; Yao and Zhang, 2002). It is known that voiced speech has harmonic structure, sinusoidal modeling (Smith and Serra, 1992) has been widely used to represent the most salient aspects of this sound, and it can be modeled as:

$$x(t)=\sum_{k=1}^{\infty}a_k(t)\exp(j(kw_0t+\alpha_k)), \tag{8}$$

where $a_k(t)$ is time-varying amplitude signal, w_0 is the initial frequency, α_k is the initial phase, k is the number of the harmonics. A key component of sinusoidal modeling is the estimation of these parameters of multiple sinusoids.

But in fact, the frequency is always changing even within a frame 10–30 ms because of the tonation. Considering the fluctuation of frequency and the harmonic structure, speech can be modeled as an AM–FM signal (Maragos et al., 1993)

$$x(t)=\sum_{k=1}^{\infty}a_k(t)\exp\left(jk\left(w_0t+\int_0^t q(\tau)\,d\tau\right)+j\alpha_k\right). \tag{9}$$

Where $q(\tau)$ is the frequency modulation function. A reasonable simplification of $q(\tau)$ is $q(\tau)=w_1t$, so we can obtain the AM–FM model (Dimitriadis and Maragos, 2005):

$$x(t)=\sum_{k=1}^{\infty}a_k(t)\exp\left(jk\left(w_0t+\frac{1}{2}w_1t^2\right)+j\alpha_k\right). \tag{10}$$

Compared the multi-component chirp signal model with AM–FM model, we can conclude that the AM–FM model also has a form of the multi-component chirp signal model. In some applications the multi-component chirp signal model is considered as a finite cut-off length of the AM–FM model. So we can make another reasonable simplification that the AM–FM speech model can be described as (Yin et al., 2008):

$$x(t)=\sum_{k=1}^{M}A_k\exp\left(jkt\left(w_0+\frac{1}{2}w_1t\right)+j\alpha_k\right). \tag{11}$$

From (11) we can find the AM–FM speech model has the multi-component chirps form. It is known that the speech signal recovery is an eternal research subject, based on this speech model, if we can estimate all the parameters of this speech model, also equates to estimate these parameters of multi-component chirps signal, then the speech is restored. Besides, this signal is not band-limit, and is non-stationary in Fourier transform domain, so the traditional parameters estimation methods in the Fourier domain are not optimal, the new signal processing tools and methods should be investigated for this kind of non-stationary signals. Base on the properties of the LCT and AM–FM speech model, two novel speech signal recovery methods in the LCT domain are proposed in the following sections.

4. Two novel recovery methods of the speech

4.1. The analysis background of speech signal

Speech can be looked as a non-stationary signal, so speech signal analysis appear very arduous. The speech signal de-noising and reconstruction is always one of the most hottest research focuses. Based on the speech sinusoidal model we review the speech signal by using the traditional methods, and then reconstruct the speech signal.

The sinusoidal model of speech can be expressed as the sum form of a group of sinusoidal wave, we hope to provide input signal $s(n)$ for K sinusoidal signal components, the original signal constitutes an approximation from sum of this K sinusoidal signal components,

$$s(n) \approx \tilde{s} = \sum_{k=1}^{K} d_k \cos(\bar{w}_k n + \phi_k), \quad n = 0, 1, 2, \ldots, n-1 \tag{12}$$

Where d_k is the signal amplitude parameter, $\bar{w}_k$ is known as the frequency parameter, ϕ_k is called the phase parameter, through such model the original signal is expressed as a group of slow-moving sine wave. The analysis phase requires to estimate all the parameters of the input signals accurately, then in the synthetic stage we can use model parameters extracted from the analysis phase to reconstruct the original speech signal. Spectral peak picking can estimate these parameters based the STFT. If the speech signal is strict periodic, then

$$s(n) = \sum_{k=1}^{K} d_k \cos(k\bar{w}_0 n + \phi_k), \quad n = 0, 1, 2, \ldots, n-1 \tag{13}$$

if the STFT of $s(n)$ is

$$S(\bar{w}) = \sum_{n=-N/2}^{N/2} s(n) \exp(-jn\bar{w}), \tag{14}$$

from the Fourier analysis the sinusoid parameters $a_k = |S(k\bar{w}_0)|$, $\bar{w}_k = k\bar{w}_0$, $\phi_k = arg(S(k\bar{w}_0))$. Through these parameters the original voice signal can be reconstructed.

But in fact, the frequency of speech is always changing even within a frame 10–30 ms because of the non-stationarity. Considering the fluctuation of frequency and the harmonic structure, speech can be modeled as an AM–FM signal from Section 3, the AM–FM speech model can be described as (18). If all the parameters of Eq. (18) can be estimated, the original voice signal will obtain the reconstruction. Below we describe how to estimate these parameters.

Maximum likelihood estimation (MLE) method of chirps signal can be expressed as (Friedlander, 1995; Abatzoglou, 1986)

$$L(a_1, a_2) = \left| \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \exp\left(-j\left(a_1 t + \frac{1}{2} a_2 t^2\right)\right) dt \right|^2, \quad -\frac{T}{2} \leqslant t \leqslant \frac{T}{2} \tag{15}$$

When L take the maximum, corresponding coordinates $(\hat{a}_1, \hat{a}_2)$ are respectively for a_1 and a_2 estimate. a_0, A are estimated as

$$\hat{a}_0 = \arg\left\{ \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \exp\left(-j\left(\hat{a}_1 t + \hat{a}_2 t^2\right)\right) dt \right\}, \tag{16}$$

$$\widehat{A} = \frac{1}{T}\left\{ \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) \exp\left(-j\left(\hat{a}_0 + \hat{a}_1 t + \hat{a}_2 t^2\right)\right) dt \right\}, \tag{17}$$

In this case, the single chirp signal parameters can be estimated, through the design of filter can also estimate the signal parameters of multi component chirps. Later appears the Polynomial Phase Transform (PPT), Chirp-Fourier Transform (CFT), Quadratic Phase Transform, Fan-Chirp Transform (FCT), and so on. They are both constantly developed tools for chirp signal analyzing and processing. These methods have their similarity.

Following the Radon-Wigner transform (RWT), Wigner-Hough transform (WHT), Radon-ambiguity transform (RAT), a later, another method is chirps signal detection and parameters estimation by using the FRFT (Qi and Tao, 2003). FRFT is a special form of the LCT. LFM signals exist maximum peak points in two-dimension plane of α and u, maximum point corresponding coordinates $\hat{\alpha}$, $\hat{u}$ can be obtained by two dimensions peak search. If we assume $X_\alpha(u)$ as the FRFT of The original chirp signal $f(t)\left(-\frac{T}{2} \leqslant t \leqslant \frac{T}{2}\right)$. They can be described as

$$\{\hat{\alpha}, \hat{u}\} = \arg|\max X_\alpha(u)|^2, \tag{18}$$

$$\hat{a}_2 = -\cot\hat{\alpha}, \quad \hat{a}_1 = \hat{u}\csc\hat{\alpha},$$

$$\hat{a}_0 = \arg\frac{X_\alpha(u)}{\sqrt{\frac{1-j\cot\hat{\alpha}}{2\pi}}\exp(j\pi\hat{u}^2\cot\hat{\alpha})}, \quad \widehat{A} = \frac{|X_\alpha(u)|}{T\left|\frac{1-j\cot\hat{\alpha}}{2\pi}\right|}. \tag{19}$$

So, it shows that we can reconstruct the original speech signal by using the estimated parameters. We put forward new methods to analyze this kind of speech signal. We introduce two methods which have more advantages compared other methods. Firstly, LCT is a relatively new transform, which used in signal processing has obvious performance, the speech signal is typically non-stationary signal, the speech signal may not band-limited when in the traditional Fourier-domain, but in the LCT domain it will appear with the good properties of band-limited, besides, Gauss noise in LCT domain shows no focusing nature, while the chirps signal have focusing nature, in addition, LCT has four parameters, which control the LCT domain properties of the speech signal, such signal properties will be more fully reflected in LCT domain. So speech signal processing is the most appropriate based on AM–FM model in LCT domain.

4.2. The LCT analysis of speech ($M = 1$)

By the definition of the LCT we know it has three free parameters, these parameters will help us to investigate the performance of the AM–FM speech in the LCT domain. From the definition of LCT as well as (11) equation, the LCT of this speech ($M = 1$) can be rewritten as

$$\begin{aligned} L^{a,b,c,d}(f(t)) &= \sqrt{\frac{1}{j2\pi b}} A \exp\left(j\left(a_0 + \frac{d}{2b}u^2\right)\right) \\ &\quad \times \int_{-\infty}^{\infty} \exp\left(\frac{-j}{b}ut\right)\exp\left(\frac{ja}{2b}t\right) \\ &\quad \times \exp\left(j\frac{1}{2}a_2t^2 + ja_1t\right)dt \\ &= \sqrt{\frac{1}{j2\pi b}} A \exp\left(j\left(a_0 + \frac{d}{2b}u^2\right)\right) \\ &\quad \times \int_{-\infty}^{\infty} \exp\left(j\frac{1}{2}\left(\frac{a}{b} + a_2\right)t^2\right) \\ &\quad \times \exp\left(jt\left(a_1 - \frac{u}{b}\right)\right)dt, \end{aligned} \tag{20}$$

if we set

$$Z_{a,b,u}(t) = \exp\left(j\frac{1}{2}\left(\frac{a}{b} + a_2\right)t^2\right)\exp\left(jt\left(a_1 - \frac{u}{b}\right)\right), \tag{21}$$

then

$$L^{a,b,c,d}(f(t)) = \sqrt{\frac{1}{j2\pi b}} A \exp\left(j\left(a_0 + \frac{d}{2b}u^2\right)\right)\int_{-\infty}^{\infty} Z_{a,b,u}(t)\,dt. \tag{22}$$

Because of the integral nature
$\int_{-T}^{T} Z_{a,b,u}(t)\,dt \leqslant \int_{-T}^{T} |Z_{a,b,u}(t)|\,dt,$
where T is sufficiently large, The largest peak just occurs if and only if $\frac{a}{b} + a_2 = 0$ and $a_1 - \frac{u}{b} = 0$.

Because now $L^{a,b,c,d}(f(t))$ includes four parameters a, b, d, u, if we want to conduct two-dimensional peak search, two parameter must be fixed. Firstly, by differential we can identify determine range of the value b, d, then respectively set the b, d value obtained, conduct respectively two-dimensional peak search on (a, u) point, we can get different peaks for different b, d values, then compare the different peaks, and select the largest peak. At this time four parameters of the corresponding the peak recorded values are $(\tilde{a}, \tilde{b}, \tilde{d}, \tilde{u})$, the estimated values of the a_2, a_1 are

$$\begin{cases} \hat{a}_2 = -\frac{\tilde{a}}{\tilde{b}} \\ \hat{a}_1 = \frac{\tilde{u}}{\tilde{b}} \end{cases}. \tag{23}$$

If we set $a = \tilde{a}$, $b = \tilde{b}$, $u = \tilde{u}$, $a_1 = \hat{a}_1$, $a_2 = \hat{a}_2$ in Eq. 21, then

$$\widetilde{Z}_{\tilde{a},\tilde{b},\tilde{u}}(t) = \exp\left(j\frac{1}{2}\left(\frac{\tilde{a}}{\tilde{b}} + \hat{a}_2\right)t^2\right)\exp\left(jt\left(\hat{a}_1 - \frac{\tilde{u}}{\tilde{b}}\right)\right) = 1, \tag{24}$$

$$|L^{\tilde{a},\tilde{b},\tilde{c},\tilde{d}}[f(t)](\tilde{u})| = \sqrt{\frac{1}{j2\pi\tilde{b}}} A\left|\int_{-\infty}^{\infty} \widetilde{Z}_{\tilde{a},\tilde{b},\tilde{u}}(t)\,dt\right|, \tag{25}$$

then

$$\begin{aligned} \frac{L^{\tilde{a},\tilde{b},\tilde{c},\tilde{d}}[f(t)](\tilde{u})}{\sqrt{\frac{1}{j2\pi\tilde{b}}}\exp\left(j\frac{\tilde{d}}{2\tilde{b}}\tilde{u}^2\right)} &= K\exp(ja_0)(K \text{ is the amplitude}) \\ &= K(\cos(a_0) + j\sin(a_0)). \end{aligned} \tag{26}$$

Let $H = \frac{L^{\tilde{a},\tilde{b},\tilde{c},\tilde{d}}[f(t)](\tilde{u})}{\sqrt{\frac{1}{j2\pi\tilde{b}}}\exp\left(j\frac{\tilde{d}}{2\tilde{b}}\tilde{u}^2\right)}$, so

$$\hat{a}_0 = arg\left\{\arctan\frac{Im(H)}{Re(H)}\right\}, \tag{27}$$

also

$$\widehat{A} = \frac{|L^{\tilde{a},\tilde{b},\tilde{c},\tilde{d}}[f(t)](\tilde{u})|}{\left|\sqrt{\frac{1}{j2\pi\tilde{b}}}\right|\left|\int_{\infty}^{\infty} \widetilde{Z}_{\tilde{a},\tilde{b},\tilde{u}}(t)\right|}. \tag{28}$$

When LCT parameters a, b, c, d reduces to $(a, b, c, d) = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha)$, then LCT becomes into the fractional Fourier transform by a fixed phase factor multiplied, then the $\hat{a}_2$, $\hat{a}_1$, $\hat{a}_0$ and $\widehat{A}$ become into the $\hat{a}_2$, $\hat{a}_1$, $\hat{a}_0$ and $\widehat{A}$ which are estimated by FRFT.

So

$$f(t) = \widehat{A}\exp\left(j\frac{1}{2}\hat{a}_2t^2 + j\hat{a}_1t + j\hat{a}_0\right), \tag{29}$$

the original speech signal ($M = 1$) can be reconstructed. Below we present two speech reconstruction method.

4.3. The first method

The first method is that the direct use of the reversibility property of the linear canonical transform,

$$L^{d,-b,-c,a}[L^{a,b,c,d}(y(t))] = y(t).$$

When the speech signal based AM–FM model carry background Gaussian noise, most of its energy concentrates in a narrow band which is as center as the peak period of LCT field, however, the background Gaussian noise in LCT domain does not have a good time–frequency focusing property (Moshinsky and Quesne, 2006; Almeida, 1994). To reconstruct the speech signal, we can use this property, that is we can design a band pass filter with peak point as the center of the filter (Qi and Tao, 2003), sample with additive noise signal through the filter can retain most energy of the speech signal and filter out most of the noise energy, then we can recover the original speech signal through the inverse LCT (Almeida, 1994).

4.4. The second method

The AM–FM model of speech is used in detection and recovery of the multi-component chirps signal. We can deal with as follows: in multi-component chirp signal detection, it occurs the strong component to impact the weak component, to avoid the impact of such interference, we can combine the first method. Firstly, set a threshold, and use it to conduct respectively four-dimensional peak search applying the quasi-Newton method, in this way we can obtain the largest peak recorded value $(\tilde{a}, \tilde{b}, \tilde{d}, \tilde{u})$, so the strong component can be detected, and we estimate its parameters as same as single chirp signal parameter estimation, and the first component can be reconstructed; secondly, to detect sub-component and we must make sure that the strong component interference can be eliminated, so we design an sequential band-pass adaptive filter in LCT domain to filter the first strong component. Then obtain the speech signal by filtered to conduct inverse LCT with parameters being $(\tilde{d}, -\tilde{b}, -\tilde{c}, \tilde{a})$, the signal becomes being in time domain. So the second chirp component can be detected, and we estimate its parameters. Repeat the above process until the magnitude of the last signal component below the predetermined threshold. Make filtering and parameter estimation combined to avoid interference effectively, we can obtain better reconstruction of the speech signal (Qi and Tao, 2003).

MLE, the discrete PPT, the discrete CFT, the QPT, and the recently proposed FCT, these chirps signal processing methods are both based on the signal peaks detection. LCT is also another method for chirps signal processing, it is also based on the signal peaks detection. Besides, conversion relationships exist among these methods.

5. Simulation results

5.1. Single chirp component of the speech

The original single chirp signal ($M = 1$) with Gaussian noise

$$y = 3\exp\left(j\left(-\frac{1}{2}8t^2 + 2t + \frac{\pi}{3}\right)\right) + w(t).$$

The SNR is 12 dB about this speech, the observation time of the original signal is $[-2\,\mathrm{s}, 2\,\mathrm{s}]$, sampling interval is $\triangle T = 0.01$ s, observations range in the LCT field $[-10, 10]$, sampling interval is $\triangle u = 0.05$. We apply the discrete LCT proposed in Aizenberg and Astola (2006), Li and Xu (2012), and Abatzoglou (1986) because of the dimensionless normalized treatment effects, these parameters must be adjusted appropriately (such as phase will occur to change) to find the correct parameter estimation, this method can refer to Koc et al. (2008).

Fig. 1 is the time domain graph of the original speech signal with Gaussian noise, and SNR of 12 dB. In Fig. 2, after fixed parameters $a = -0.99211$, $b = 0.12533$, $d = -0.99211$ a peak value is searched. In Fig. 3, set

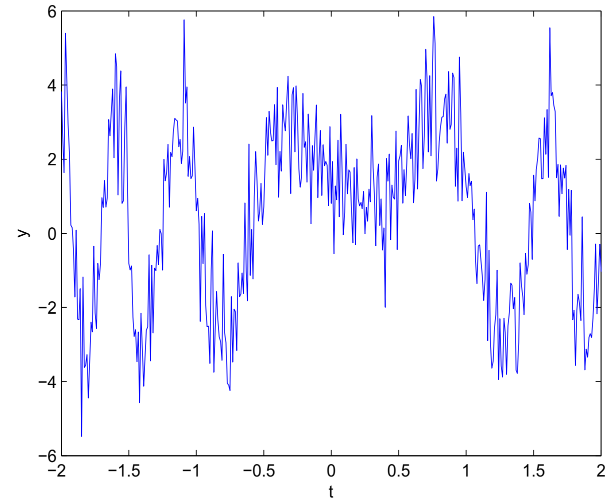

Fig. 1. The original chirp signal needed to restore with noise.

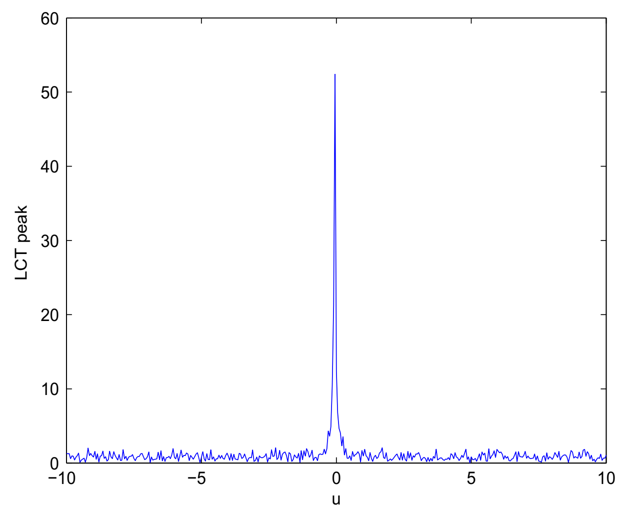

Fig. 2. Set $a = -0.99211$, $b = 0.12533$, $d = -0.99211$, search for the peak after the LCT.

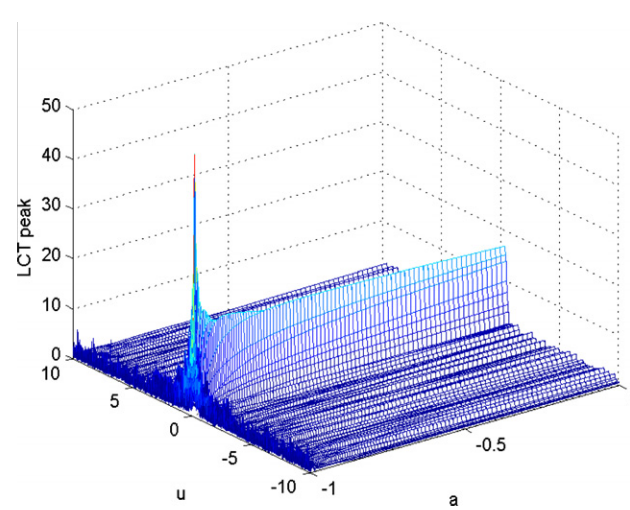

Fig. 3. Set $b = 0.12533$, $d = -0.99211$, exists the peak point after LCT.

$b = 0.12533$, $d = -0.99211$, the graph show the spectrum of the chirp signal in LCT domain, from the graph we can see the peak value, quasi-Newton method can be used as four-dimensional peak search. Then we can find the corresponding coordinates $(\tilde{a}, \tilde{b}, \tilde{d}, \tilde{u})$. By the parameters estimation modulation frequency, angular frequency, phase and amplitude respectively are $\hat{a}_2 = -7.9158$, $\hat{a}_1 = 1.9947$, $\hat{a}_0 = 1.0071$, $\widehat{A} = 3.0619$. Fig. 4 shows the first method to restore the original speech signal. Fig. 5 shows comparisons between the original chirp signal and the reconstructed signal by the parameters estimation in the other four kinds of transforms domain. Fig. 6 shows mean error rate comparisons of these methods in amplitude between the proposed method and other methods. From Fig. 6, we can see that mean error rates of these methods are similar during [0s,1s], differences are not very significant, but during [−2 s, 0 s] and [1 s, 2 s], mean error rates by ML method is the most significant in these methods, then is PPT method and Dechirp method, mean error rate by LCT method is not the most effective. From Fig. 6, it shows that LCT is feasible in speech signal recovery compared with other methods.

5.2. Speech of multi-component chirps for $M = 2$

The speech signal ($M = 2$) with Gaussian noise

$$y = 3\exp\left(j\left(-\frac{1}{2}8t^2 + 2t + \frac{\pi}{4}\right)\right) + 4\exp\left(j\left(-\frac{1}{2}18t^2 + 5t + \frac{\pi}{3}\right)\right) + w(t).$$

The SNR is 22 dB about this speech, the observation time of the original signal is [−2 s, 2 s], sampling interval is $\triangle T = 0.01$ s, observations range in the LCT field [−10, 10], sampling interval is $\triangle u = 0.05$.

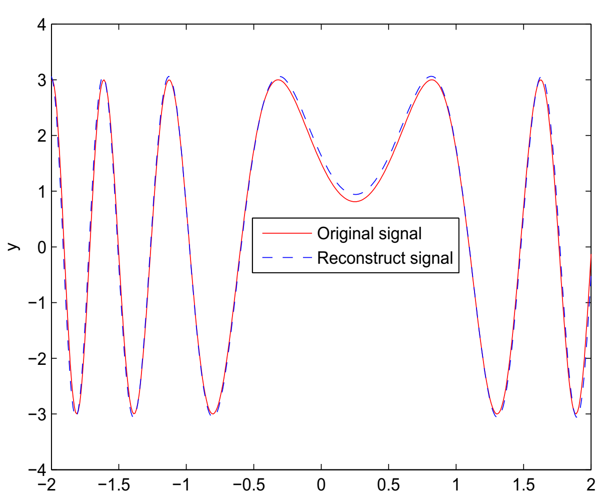

Fig. 4. Reconstruction of signal by the inverse LCT compared with the original signal.

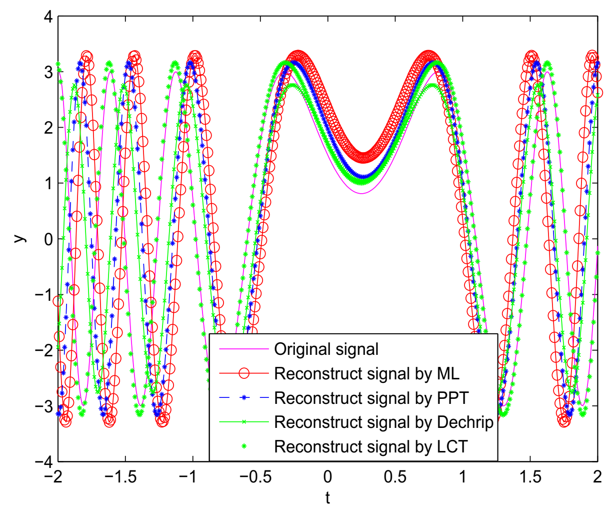

Fig. 5. Reconstruct the signal by using parameter estimation compared with the original speech.

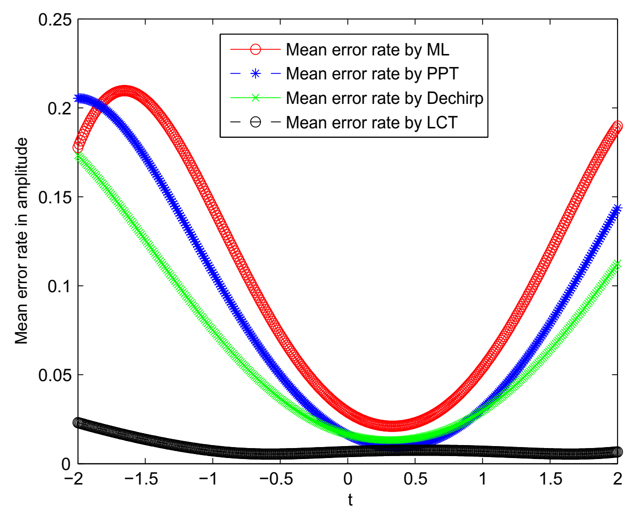

Fig. 6. Mean error rate in amplitude using parameters estimation by LCT compared with other methods when SNR of 12 dB.

Fig. 7 is the time domain graph of the speech signal ($M = 2$) with Gaussian noise, and SNR of 22 dB. From Fig. 8 we identified two peaks after a four-dimensional peaks search on (a, u) plane. Fig. 9 shows search the second peak after inhibiting the first component. Fig. 10 shows the first method to restore the speech signal. By the parameters estimation modulation frequency, angular frequency, phase and amplitude were respectively $\hat{a}_1 = -7.9011$, $\hat{b}_1 = 2.1047$, $\hat{c}_1 = 0.7480$, $\widehat{A}_1 = 2.9021$, $\hat{a}_2 = -18.1348$, $\hat{b}_2 = 4.9024$, $\hat{c}_2 = 1.0923$, $\widehat{A}_2 = 4.1262$. Fig. 11 shows comparisons between the original chirp signal and the reconstructed signal by the parameters estimation in LCT domain and in the other three kinds of transforms domain. Fig. 12 shows mean error rate comparisons of these methods in amplitude between the proposed method and other methods. From Fig. 12, we can see that mean error rates of these methods are similar during [−0.5 s, 1 s], differences

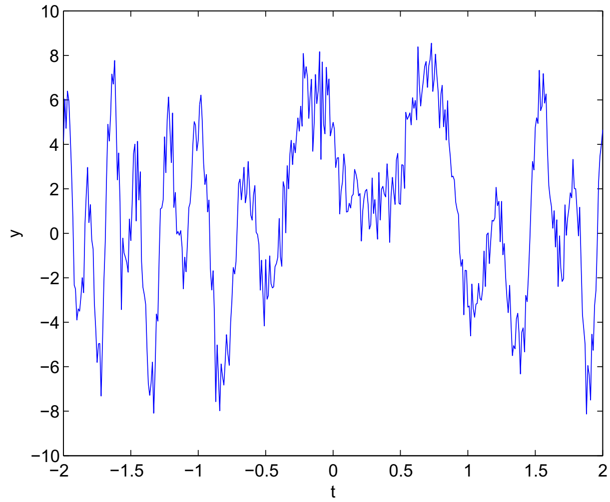

Fig. 7. The original speech signal.

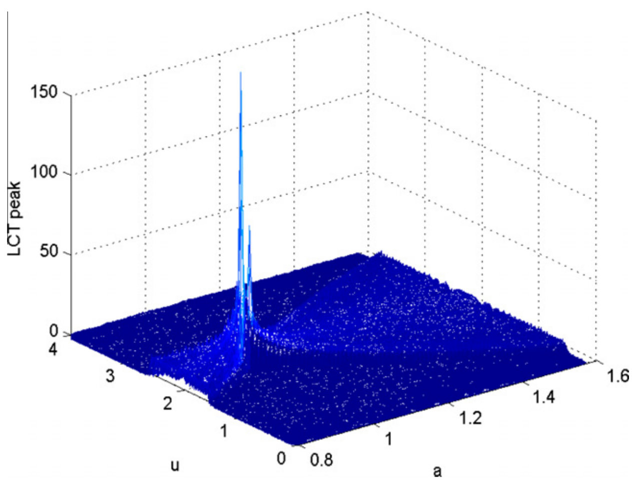

Fig. 8. Search to peaks of the two components.

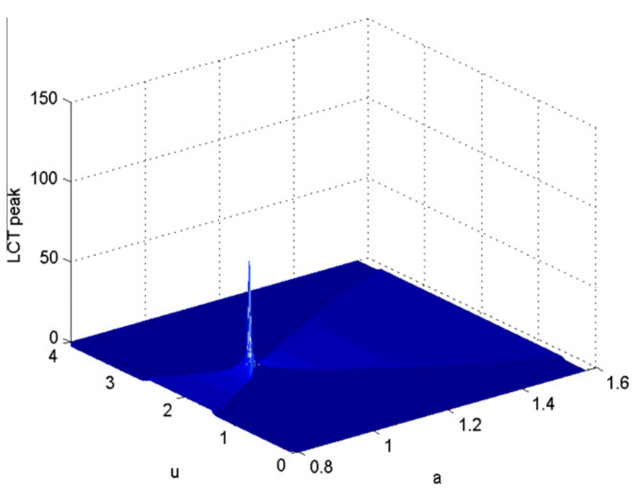

Fig. 9. Inhibit the first component to search the second peak.

are not very significant, but during other interval areas, mean error rate by ML method is the most significant in

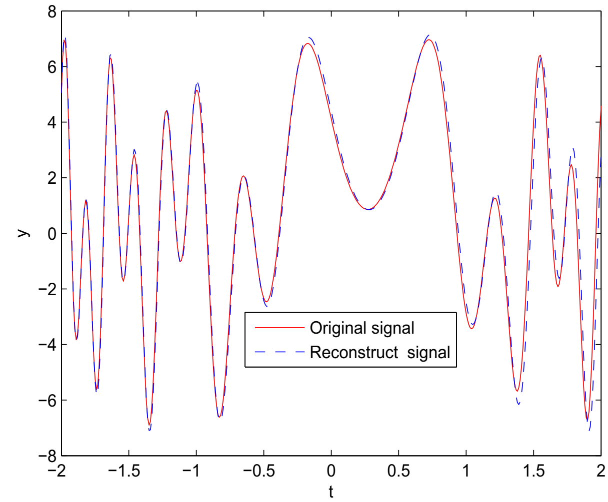

Fig. 10. Reconstruction of speech by the first method.

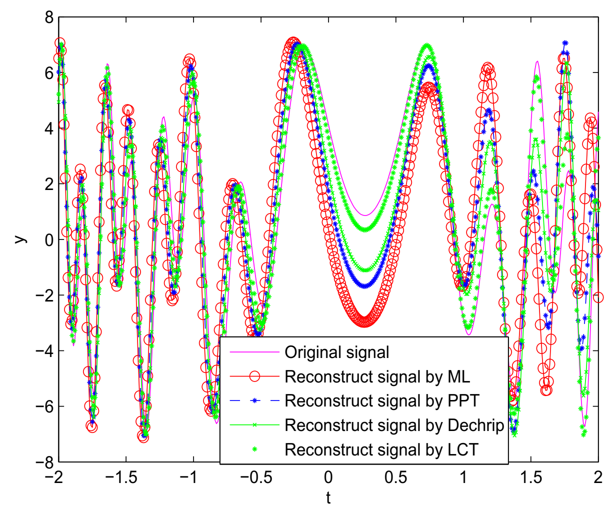

Fig. 11. Reconstruct the speech by using parameter estimation compared with the original speech.

these methods, then is PPT method and Dechirp method, mean error rate by LCT method is not the most significant. From Fig. 12, it shows that LCT is feasible in speech signal recovery compared with other methods.

5.3. The experiment of real speech signal

From the references, we know that speech, which can be considered as the AM–FM model, is a special case of the multicomponent chirp signals. It is shown in Section 5.2 that we has done on artificial structure signal for experimental analysis, now we use real speech signals for experimental analysis. We collected source speech “we” with a sampling rate of 44 kHz, and SNR of 40 dB by means of a notebook computer with recorder. Fig. 13 is the time domain graphic of the source speech “we” with Gaussian noise, and SNR of 40 dB. Fig. 14 shows the first method to restore the source speech “we”, mean error rate of recon-

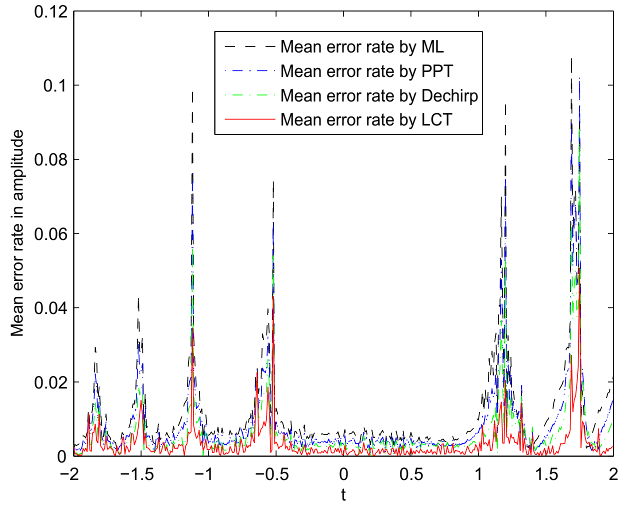

Fig. 12. Mean error rate in amplitude using parameters estimation by LCT compared with other methods when SNR of 22 dB.

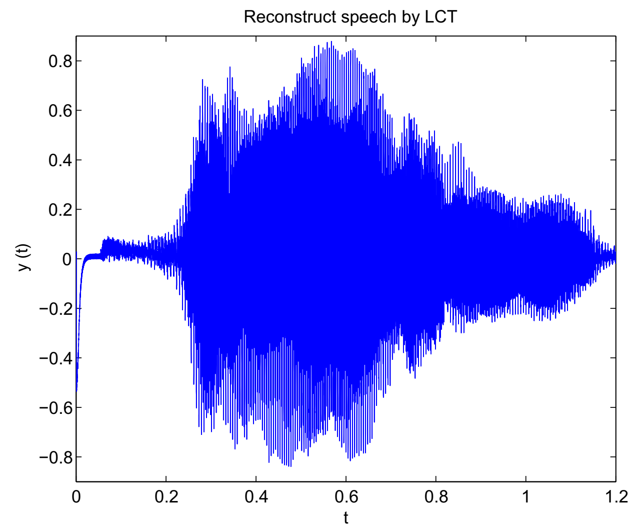

Fig. 15. Reconstruct the speech “we” by using parameter estimation.

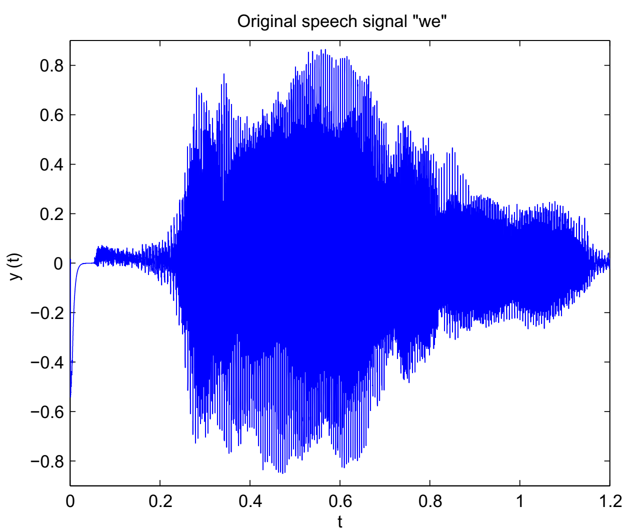

Fig. 13. Source speech signal “we”.

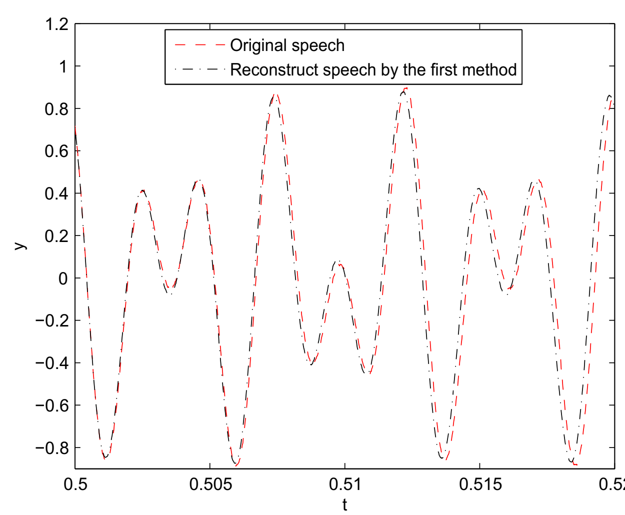

Fig. 16. Reconstruct the speech “we” between 0.5 s and 0.52 s by using the first method.

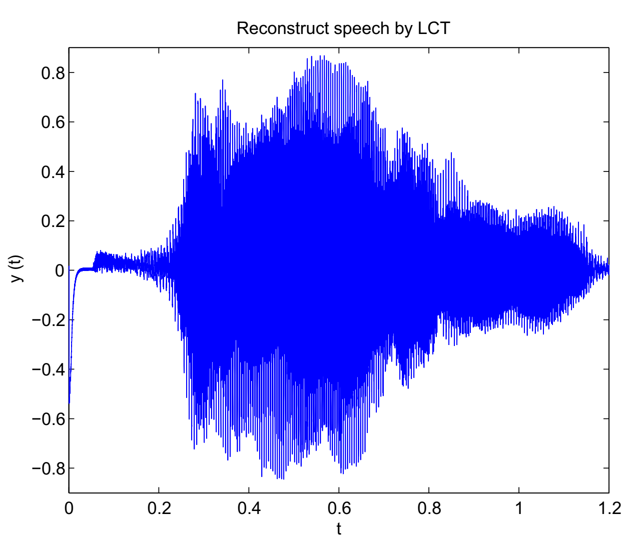

Fig. 14. Reconstruction of speech “we” by the first method.

structed speech is 0.1520. Fig. 15 shows the reconstructed speech signal “we” by the parameters estimation in LCT domain, mean error rate of reconstructed speech is 0.1460. In general, the “M” cannot be estimated perfectly, we can obtain different results for if we use the different threshold values. for example, when the threshold is set to be 0.2, by means of the second method the order number of chirp components “M” is estimated to 21, however, the perfect estimation method and how to choose the right threshold will be one of our research directions in the future.

In order to see the differences between the original and reconstructed signal more clearly, Fig. 16 gives the comparison of amplification between the original and reconstructed signals during time 0.5 and 0.52 s by using the second method. From Fig. 16 we can see that the fundamental frequency of this region is $1/0.0075\ (\mathrm{s}) = 130$ Hz,

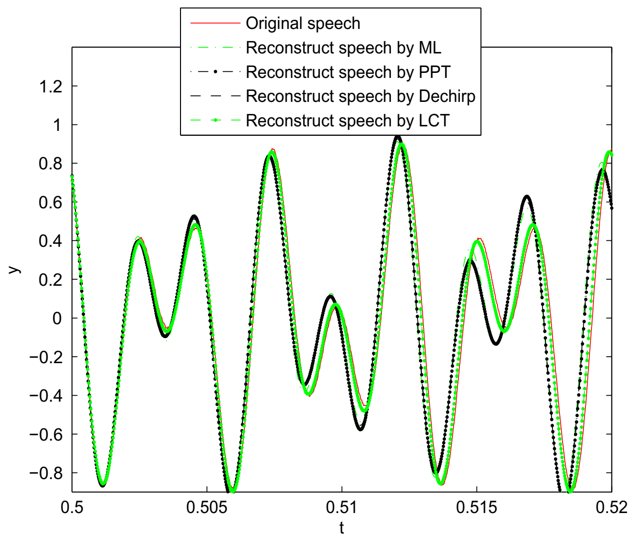

Fig. 17. Reconstruct the speech "we" between 0.5 s and 0.52 s by using parameter estimation compared with the original speech.

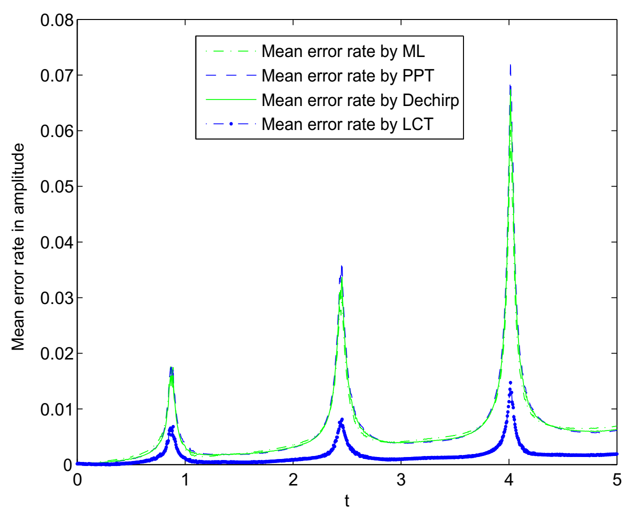

Fig. 18. Mean error rate of reconstruct the speech "we" between 0.5 s and 0.52 s by LCT compared with other methods when SNR of 40 dB.

because that is a real voiced speech segment, the Fig. 16 should includes at least 2.5 cycles in this 0.02 s segment, this results verify the correctness of the methods proposed in this paper.

Fig. 17 shows comparisons between the original speech and the reconstructed speech by the parameters estimation in LCT domain and in the other three kinds of transforms domain. Fig. 18 shows mean error rate comparisons of these methods in amplitude between the proposed method and other methods. In Fig. 18, mean error rate by PPT method is the most significant in these methods, then is Dechirp method and ML method, mean error rate by LCT method is little compared to other methods. This results shows that the proposed method is also effective for real speech signal recovery.

6. Conclusion

In this paper two recovery methods of speech based on the AM–FM model in the LCT domain are proposed. One depends on the linear canonical transform domain filtering, the other is based signal parameter estimation to restore the speech signal. The experiments are also proposed to show the performance of the methods. However, all had their own shortcomings, filtering out the noise signal energy maybe inadequate in these methods. Besides, in multi-component chirp signal detection, it occurs the strong component to impact the weak component, to avoid the impact of such interference, these problems will be investigated in detail in our future works. At the same time, there certainly exist some problems about the real applications of the proposed method in real speech signal processing, for example, the ideal order number of chirp components "M" in real speech signal, in general, cannot be estimated perfectly, this problem should be investigated and discussed in future works.

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